Sleepmaxxing: The Science Behind a Viral Sleep Optimization Trend

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Abstract

Sleepmaxxing has emerged as a heterogeneous approach that has gained increasing visibility through social media and digital health culture, aiming to optimize sleep duration, sleep quality, sleep regularity, and environmental conditions through various behavioral, technological, and biological strategies. This narrative review aimed to evaluate sleepmaxxing practices in light of current scientific evidence, distinguish evidence-supported approaches from those with limited evidence, and discuss the potential clinical and public health implications of this trend. A literature search was conducted in the PubMed database, with particular emphasis on studies published between 2021 and 2026, alongside earlier studies considered fundamental to the topic. Current evidence indicates that adequate and regular sleep, consistent sleep-wake schedules, appropriate physical activity, healthy dietary habits, and cognitive behavioral therapy for insomnia when clinically indicated have scientific support for promoting sleep health. In contrast, evidence regarding blue-light-blocking glasses, melatonin and magnesium supplementation, white noise, mouth and nasal taping, consumer sleep trackers, and smart beds remains limited, heterogeneous, or inconsistent. Furthermore, excessive sleep monitoring and the pursuit of “perfect sleep” may increase sleep-related anxiety and contribute to orthosomnia-like behaviors. The presentation of scientifically unsupported or weakly supported practices as definitive solutions on social media also represents an important public health concern. Overall, sleepmaxxing should not be considered a scientifically validated, standardized sleep optimization protocol, but rather a digital sleep culture encompassing practices with varying levels of scientific evidence. Future research should focus on developing valid tools to directly measure sleepmaxxing behaviors, evaluating the accuracy of wearable sleep technologies, and investigating their long-term clinical and psychological consequences.

Keywords: Sleepmaxxing; sleep health; sleep optimization; sleep hygiene; sleep regularity; sleep tracking; orthosomnia; social media

1. Introduction

Sleep is one of the fundamental biological processes essential for maintaining physical and mental health and plays a critical role in neurocognitive function, metabolic homeostasis, immune system regulation, and cardiovascular function. Adequate sleep duration and quality re essential for learning, memory, emotional regulation, and daily performance, whereas chronic sleep deprivation has been associated with obesity, type 2 diabetes mellitus, hypertension, depression, and cardiovascular diseases. Chronic stress, irregular working schedules, and increasing exposure to digital screens associated with modern lifestyles negatively affect sleep quality, making approaches aimed at improving sleep health a major focus of both scientific research and public interest.

One of the most notable reflections of this growing interest is the emergence of the concept of *sleepmaxxing*, which has gained prominence alongside the biohacking movement aimed at optimizing biological performance. Sleepmaxxing is a novel lifestyle approach that refers to the systematic implementation of various behavioral, environmental, and technological strategies to improve sleep duration, quality, and depth. Derived from the *looksmaxxing* trend, the term has evolved into an umbrella concept encompassing numerous practices, including blue light-blocking glasses, mouth taping, magnesium and melatonin supplementation, white noise applications, sleep-tracking devices, and various nighttime routines. These principal domains of sleepmaxxing practices are conceptually summarized in Figure 1.

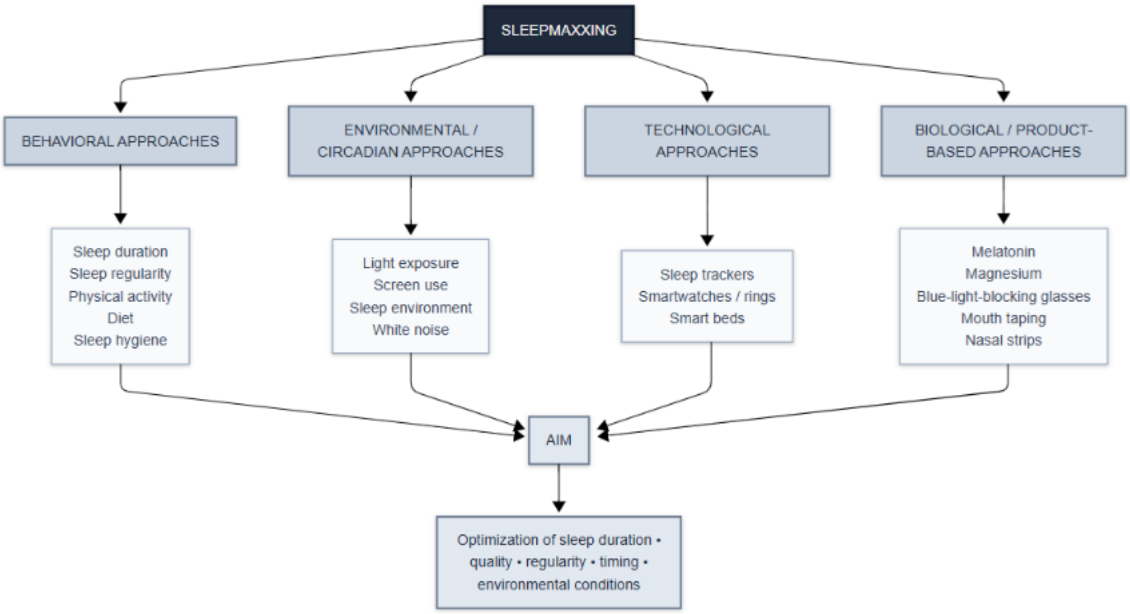


Figure 1. Conceptual framework of the principal domains of sleepmaxxing practices.

Sleepmaxxing has gained global popularity largely through social media platforms, particularly TikTok, as well as Instagram, YouTube, and Reddit. Among younger users in particular, content garnering millions of views has portrayed sleep optimization not merely as a component of healthy living but also as an indicator of high performance and productivity. At the same time, the widespread adoption of wearable technologies has enabled individuals to monitor their sleep stages and sleep scores, reinforcing the perception of sleep as a measurable performance metric that can be continuously optimized.

However, the scientific evidence supporting the practices promoted under the concept of sleepmaxxing is heterogeneous. While some approaches, such as improving sleep hygiene, engaging in regular physical activity, and supporting circadian rhythm, are backed by strong scientific evidence, the evidence for mouth taping, various dietary supplements, and many commercial sleep products remains limited or conflicting. Furthermore, the rapid dissemination of unverified health information on social media can blur the distinction between evidence-based recommendations and speculative practices, potentially encouraging the pursuit of "perfect sleep" and contributing to the emergence of new behavioral concerns such as orthosomnia.

The aim of this narrative review is to evaluate the concept of sleepmaxxing, which has become increasingly viral on social media in recent years, in light of the current scientific literature; to examine the level of evidence supporting the practices promoted within this approach, compare scientifically supported and unsupported strategies, and discuss the potential implications of sleepmaxxing for clinical practice and public health.

2. Literature Search Strategy

In this narrative review, a literature search was conducted in the PubMed database to evaluate the current scientific evidence regarding the concept of *sleepmaxxing* and the sleep optimization practices promoted within this framework. Studies published during the past five years (2021–2026) were prioritized. However, earlier landmark studies considered essential for explaining the underlying concepts and interpreting current evidence were also included. All figures and graphs presented in this study were originally developed by the author to synthesize and visualize the scientific evidence identified through the literature search.

The literature search employed the keywords *"sleepmaxxing," "sleep optimization," "sleep hygiene," "sleep quality," "sleep duration," "circadian rhythm," "light exposure," "physical activity and sleep," "melatonin," "magnesium and sleep," "mouth taping," "blue light blocking glasses," "wearable sleep trackers," "white noise,"* and *"orthosomnia"*, as well as appropriate combinations of these terms. In addition, systematic reviews, meta-analyses, randomized controlled trials, observational studies, and recent clinical guidelines addressing the practices commonly promoted within the sleepmaxxing framework were prioritized for inclusion.

Studies were evaluated according to the following primary criteria: direct relevance to sleep health, sleep quality, or sleep duration; reliance on human research; publication in peer-reviewed scientific journals; and provision of current evidence relevant to the research question. Studies not directly related to the topic, investigations based solely on animal models, and publications lacking sufficient scientific evidence were excluded from the review. The findings were narratively synthesized to comparatively evaluate the current level of scientific evidence, as well as the potential benefits and risks, of the practices promoted within the sleepmaxxing framework.

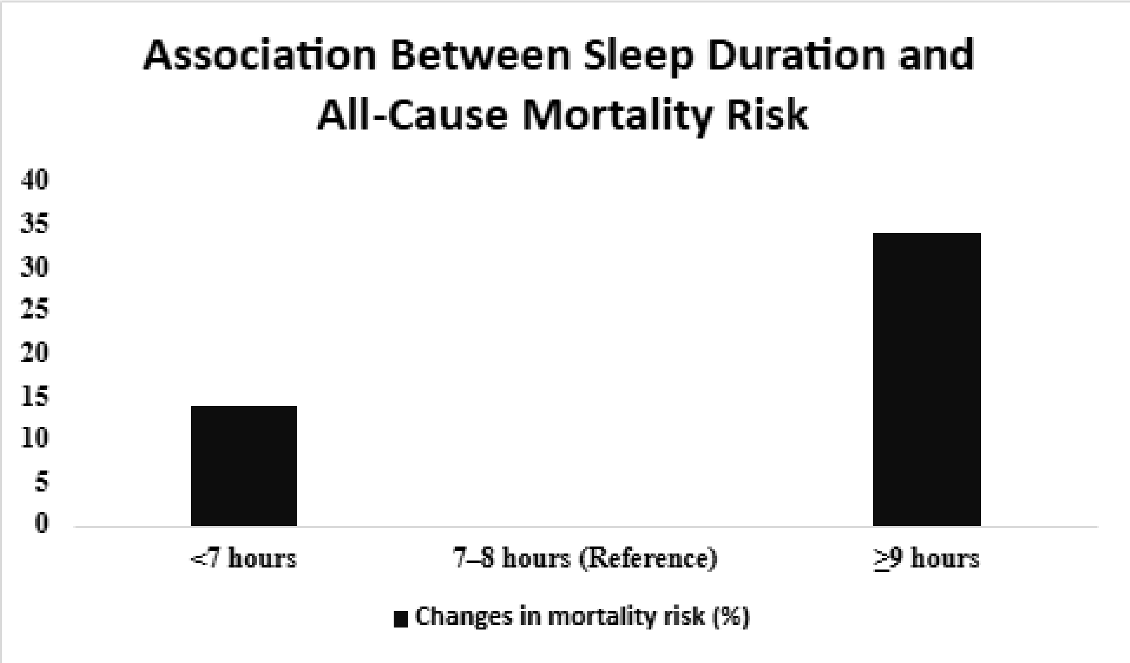
3. What Is Currently Known About Sleepmaxxing?

Although many practices promoted under the concept of sleepmaxxing are presented as a new social media trend, a substantial proportion of them are actually based on behavioral and environmental approaches that have long been the subject of sleep science research. The current literature indicates that sleep health should be evaluated not only in terms of total sleep duration but also through multiple components, including sleep regularity, sleep timing, environmental conditions, physical activity, and nutrition. Therefore, the scientific evaluation of sleepmaxxing practices should be based on the quality and strength of the available evidence rather than on their popularity.

3.1. Sleep Duration and Regularity

One of the fundamental components of sleep optimization is maintaining adequate and consistent sleep habits. A comprehensive review has shown that both short and long sleep durations are associated with a variety of adverse health outcomes, including cardiovascular disease, cognitive decline, depression, metabolic syndrome, and stroke. The finding that each one-hour reduction in sleep duration is associated with an increased risk of certain health outcomes further underscores the importance of obtaining sufficient sleep. [5] A more recent meta-analysis reported that sleeping fewer than 7 hours per night was associated with a 14% increase in all-cause mortality compared with sleeping 7–8 hours, whereas sleeping 9 hours or more was associated with a 34% increase in all-cause mortality. [7] However, these findings do not indicate that prolonged sleep duration is a direct cause of disease; rather, they suggest that the relationship between sleep duration and health outcomes is complex. [5,7] This association between sleep duration and all-cause mortality is visually summarized in Graph 1.

In addition to sleep duration, the regularity of sleep and wake times has increasingly been recognized as an important indicator of sleep health. A recent systematic review found that irregular sleep patterns are associated with depressive and anxiety symptoms, higher body mass index, insulin resistance, hypertension, and cardiovascular events. Furthermore, irregular sleep has been linked to reduced hippocampal volume and an increased risk of dementia. [1] These findings suggest that, within the sleepmaxxing approach, the goal should not be limited to simply “getting more sleep,” but should also emphasize maintaining a consistent sleep–wake rhythm. Therefore, sleep optimization strategies should be considered a multidimensional process that integrates both quantitative and qualitative aspects of sleep. Beyond increasing total sleep time, improving sleep regularity and stability may represent a critical component of achieving sustainable sleep health and supporting the potential benefits of sleep optimization practices.



Graph 1. Relationship between sleep duration and the risk of all-cause mortality.

Note: The findings indicate an association and should not be interpreted as evidence of causality.

3.2 Circadian Rhythm and Light Exposure

Light exposure is an important environmental factor in the regulation of the sleep–wake cycle. In particular, both the intensity and timing of light exposure are thought to influence the circadian system. However, a systematic review examining personal light exposure under real-life conditions in healthy adults reported that the relationship between light exposure and the sleep–wake rhythm and sleep characteristics was not consistent across all outcomes. While limited evidence suggested positive associations between the amount and timing of light exposure and the rest–activity rhythm, as well as certain indicators of sleep architecture, the evidence regarding sleep quality, sleep estimates, circadian phase, and mood was found to be inconsistent. [2]

Therefore, although regulating light exposure is a biologically plausible strategy within the sleepmaxxing approach, the current evidence does not demonstrate that this intervention consistently and directly improves sleep quality in every individual. In particular, the predominance of cross-sectional studies assessing personal light exposure under real-world conditions, together with the limited number of interventional studies, highlights the need for more robust research in this area. [2]

3.3 Physical Activity

Physical activity is another behavioral strategy commonly promoted within the sleepmaxxing approach to improve sleep quality. A meta-analysis examining diverse populations reported that, although the overall effect of physical activity on sleep quality was not statistically significant, low- and moderate-intensity physical activity was associated with significant improvements in sleep quality. The effect varied across age groups, being more pronounced in children and middle-aged and older adults, whereas no significant effect was observed in young adults. Therefore, although physical activity represents a potentially beneficial lifestyle component for sleep health, it should not be generalized within the context of sleepmaxxing that “more intense exercise equals better sleep.” [3]

3.4 Nutrition

Dietary habits are also considered modifiable lifestyle factors that may influence sleep characteristics. A systematic review evaluating the association between adherence to the Mediterranean diet and sleep outcomes reported that greater adherence to the Mediterranean diet was associated with better sleep quality, adequate sleep duration, and lower levels of daytime sleepiness and insomnia symptoms. However, because the available evidence is largely based on observational studies, a causal relationship cannot be established with certainty. Furthermore, the relationship between diet and sleep is likely bidirectional, highlighting the need for further interventional studies to better clarify this association. [4]

From this perspective, rather than promoting a specific food or dietary supplement as a definitive method for “optimizing sleep” within the sleepmaxxing approach, it appears more evidence-based to consider overall dietary quality as an integral component of sleep health. [4]

3.5 Insomnia and Cognitive Behavioral Therapy

Among the strategies promoted within the sleepmaxxing approach, cognitive behavioral therapy for insomnia (CBT-I) has the strongest clinical evidence base. A systematic review and meta-analysis examining insomnia in individuals with coexisting mental health disorders demonstrated that CBT-I produces moderate to large reductions in insomnia severity. Significant improvements in insomnia symptoms were also reported across a range of comorbid conditions, including depression, post-traumatic stress disorder, and alcohol use disorder. [6]

These findings indicate that sleepmaxxing is not merely a collection of products and “sleep hacks” popularized on social media; rather, scientifically validated behavioral interventions play a far more substantial role in the management of sleep-related clinical conditions. For individuals experiencing insomnia, it is therefore more important to appropriately assess the underlying sleep disorder and, when indicated, utilize evidence-based treatment approaches than to focus solely on increasing sleep duration or improving sleep scores. [6]

Overall, the current evidence suggests that several components commonly associated with sleepmaxxing—including maintaining regular sleep–wake schedules, obtaining sufficient sleep duration, engaging in appropriate physical activity, adopting healthy dietary habits, and using CBT-I for insomnia—are supported by scientific evidence. [1,3–7] However, in certain areas, such as light exposure, the available evidence remains limited or heterogeneous, indicating that not every sleep optimization strategy promoted on social media is supported by the same level of scientific evidence. [2] Therefore, the evaluation of sleepmaxxing practices should be based on the quality of the available scientific evidence and their clinical applicability rather than on their popularity.

4. Which Sleepmaxxing Practices for Improving Sleep Are Scientifically Supported and Which Are Not?

The practices promoted under the sleepmaxxing trend on social media and digital health platforms are highly heterogeneous in terms of the strength of their scientific evidence. Although some interventions are based on biologically plausible mechanisms related to sleep physiology, these mechanisms have not always been shown to translate into clinically meaningful improvements in sleep. In particular, the current evidence for interventions such as blue light-blocking glasses, mouth taping, magnesium and melatonin supplementation, white noise, nasal strips, and consumer-grade sleep trackers ranges from strong and consistent support to limited or conflicting findings. Therefore, the evaluation of sleepmaxxing practices should not rely solely on the existence of a biologically plausible mechanism but should also consider the results of randomized controlled trials, sample sizes, objective sleep measurements, and the consistency of findings across studies. [8–21] The evidence hierarchy used in this evaluation is presented in Figure 2.

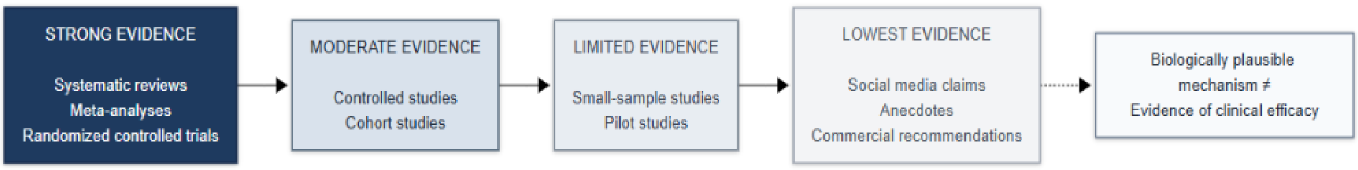


Figure 2. Evidence hierarchy used for the evaluation of sleepmaxxing practices.

4.1. Blue Light-Blocking Glasses

Blue light-blocking glasses are a non-pharmacological sleep intervention designed to reduce retinal exposure to short-wavelength light, particularly during the evening hours. The theoretical basis of this approach is that evening exposure to blue light may activate intrinsically photosensitive retinal ganglion cells (ipRGCs), leading to melatonin suppression and a delay in the circadian rhythm. Consequently, reducing blue light exposure has been proposed as a strategy to facilitate sleep onset. [9]

However, the existence of a plausible biological mechanism does not necessarily imply that clinical efficacy has been definitively established. A 2025 systematic review and meta-analysis that included only randomized crossover trials reporting actigraphic outcomes evaluated three double-blind RCTs involving a total of 49 participants. Although the use of blue light-blocking glasses reduced sleep onset latency by an average of 4.86 minutes and increased total sleep time by 8.75 minutes, neither outcome reached statistical significance. Similarly, no significant differences were observed in sleep efficiency or wake after sleep onset. [8]

On the other hand, a broader systematic review evaluating blue light-blocking glasses across various conditions—including sleep disorders, jet lag, and shift work—analyzed 24 publications and reported that several studies demonstrated favorable effects, particularly in reducing sleep onset latency. However, differences in study methodologies and participant populations limited the generalizability of these findings. [9]

Therefore, the current evidence suggests that blue light-blocking glasses represent a biologically plausible but not yet definitively validated sleepmaxxing strategy. In particular, the failure of recent randomized controlled trials using objective actigraphic outcomes to demonstrate significant benefits indicates that this intervention should be interpreted cautiously rather than promoted as a method that definitively improves sleep. [8,9]

4.2. Mouth Taping

Mouth taping is a practice that involves sealing the lips with adhesive tape during sleep to reduce mouth breathing and promote nasal breathing. On social media, this method has been associated with a wide range of claimed benefits, including improved sleep, reduced snoring, better oral health, and even increased energy levels. However, many of these claims have not been directly evaluated in clinical research. [10]

A comprehensive literature review on mouth taping identified 177 unique studies, but only nine met the inclusion criteria. Some of these studies reported improvements in obstructive sleep apnea (OSA) parameters, snoring, or oral air leakage during non-invasive ventilation. Nevertheless, due to the heterogeneity of the available studies, no strong consensus regarding the overall effectiveness of mouth taping could be reached. The review also emphasized that only a small proportion of the claims promoted on social media have been scientifically investigated. [10]

In addition, a preliminary study involving 20 mouth-breathing individuals with mild OSA found that mouth taping during sleep reduced the median apnea–hypopnea index from 8.3 to 4.7 events per hour and decreased the snoring index by approximately 47%. Improvements in the oxygen desaturation index were also observed. However, this was a small pilot study limited to a specific patient population; therefore, its findings cannot be directly generalized to healthy individuals or the broader sleepmaxxing population. [11]

Accordingly, although mouth taping has shown potential benefits in selected patients with OSA and snoring, there is currently insufficient evidence to consider it a scientifically validated sleepmaxxing strategy for improving sleep in the general population. The broad health claims commonly promoted on social media should therefore be distinguished from the limited evidence available from clinical research. [10,11]

4.3. Magnesium Supplementation

Magnesium is one of the most popular supplements promoted within the sleepmaxxing movement to improve sleep quality and reduce insomnia. Owing to its role in neuromuscular and neurological function, magnesium has been suggested to influence sleep regulation. However, the clinical evidence regarding its effects on sleep differs between observational studies and randomized controlled trials. [12]

A systematic review investigating the relationship between magnesium and sleep health evaluated nine cross-sectional, cohort, and randomized controlled studies involving a total of 7,582 participants. While observational studies suggested an association between magnesium status and sleep quality, randomized controlled trials produced conflicting findings regarding the effects of magnesium supplementation on sleep disorders. Consequently, although the available evidence indicates a relationship between magnesium status and sleep quality, it does not establish that this association is directly or causally attributable to magnesium supplementation. [12]

A more recent systematic review found that, among eight intervention studies evaluating sleep-related outcomes, five reported improvements in at least one sleep parameter, two reported no significant improvements, and one yielded mixed results. However, interpretation of these findings is complicated by generally small sample sizes, variations in magnesium dosage and formulation, and the inclusion of additional active ingredients in some interventions. [13]

Taken together, these findings suggest that magnesium may have potential as a sleep-supporting intervention; however, the current evidence is not sufficiently strong to conclude that routine magnesium supplementation improves sleep in the general population. In particular, the widespread portrayal of magnesium on social media as being "essential for sleep" or a "direct treatment for insomnia" extends beyond the available clinical evidence. [12,13]

4.4. Melatonin

Melatonin is one of the supplements within the sleepmaxxing approach with the strongest scientific foundation; however, its effectiveness varies considerably depending on the indication for use and the target population. Due to its role in regulating the circadian rhythm, melatonin is believed to be particularly effective in influencing sleep timing and facilitating sleep onset. [14,15]

A meta-analysis of 23 randomized controlled trials involving adults with various health conditions reported that melatonin supplementation resulted in a statistically significant improvement in sleep quality as assessed by the Pittsburgh Sleep Quality Index. However, the heterogeneity among the included studies was substantial ($I^2 = 80.7\%$), and the effects of melatonin differed across disease groups. Consequently, the observed benefits cannot be readily generalized to all adult populations. [14]

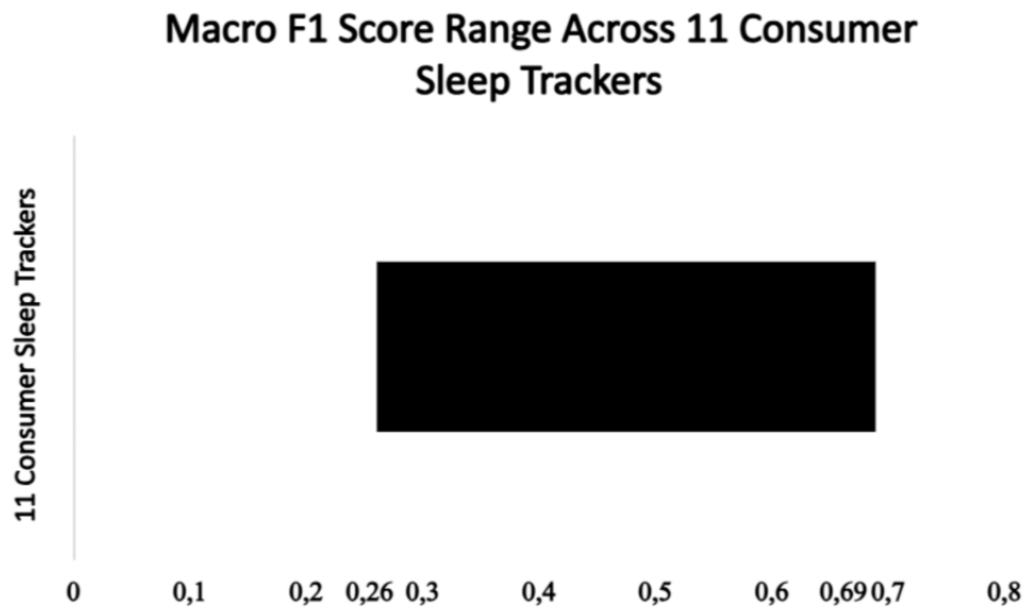
A more comprehensive systematic review and meta-analysis focusing on chronic insomnia reached somewhat different conclusions. In adults with chronic insomnia and no coexisting medical conditions, melatonin did not produce significant improvements in sleep onset latency, total sleep time, or sleep efficiency. In contrast, more favorable outcomes were observed in children and adolescents, as well as in populations with certain comorbid conditions. [15]

Therefore, the current evidence suggests that melatonin is not an unsupported intervention; rather, it may be effective in specific clinical contexts but should not be regarded as a universal "sleep enhancer" for everyone within the context of sleepmaxxing. In particular, the evidence supporting its use in adults with chronic insomnia remains limited and highly context-dependent. [14,15]

4.5. Consumer Sleep Trackers and Wearable Technologies

Smartwatches, smart rings, sleep mats, and other consumer sleep tracking devices constitute a major component of the sleepmaxxing philosophy of "optimizing sleep through measurement." However, when evaluating these devices, two distinct questions should be considered separately: How accurately does the device measure sleep, and has the use of this information actually been shown to improve users' sleep? [16,17]

In a multicenter validation study involving data from 75 participants, the agreement of 11 different consumer sleep tracking devices with polysomnography was evaluated. The results demonstrated considerable variability among devices in their ability to classify sleep stages, with macro F1 scores ranging from 0.26 to 0.69. While some devices performed better in identifying specific sleep stages, others showed only partial agreement with polysomnography. [16] The variability in performance among these devices is illustrated in Graph 2.



Graph 2. Variability in reported Macro F1 scores and sleep stage classification performance across 11 consumer sleep tracking devices.

The scientific position statement evaluating the use of wearable technologies in sleep and circadian rhythm research also concludes that consumer-grade devices have considerable potential but are subject to important limitations, including the misclassification of wakefulness, the inability to accurately measure sleep occurring outside the main sleep period, susceptibility to artifacts, and variability in performance across individuals with different characteristics. Furthermore, the performance of these devices may be influenced by certain personal characteristics, raising important considerations regarding health equity. [17]

Therefore, consumer sleep tracking devices may serve as useful tools for assessing sleep but should not be regarded as clinically validated sleep treatments. The ability of a device to measure a sleep parameter does not imply that it can improve that parameter. This distinction is particularly important in the context of sleepmaxxing, as users may incorrectly interpret device-generated sleep scores as direct indicators of “good” or “bad” sleep. [16,17]

4.6. White Noise

White noise is a non-pharmacological intervention that is thought to facilitate sleep onset and sleep maintenance by masking environmental sounds and reducing the impact of external stimuli. A recent systematic review and meta-analysis evaluated 12 randomized controlled trials involving 1,301 participants and found that the effects of white noise varied according to age group and clinical setting. [18]

Among adults and older adults, white noise was associated with a significant reduction in Pittsburgh Sleep Quality Index (PSQI) scores, indicating an improvement in subjective sleep quality. Significant reductions in PSQI scores were also observed in both intensive care unit (ICU) and non-ICU clinical settings. However, no consistent benefits were demonstrated across all groups for objective sleep outcomes such as total sleep time, sleep efficiency, or wake after sleep onset. [18]

Therefore, compared with several other sleepmaxxing practices, white noise may be considered an intervention supported by relatively stronger—although still heterogeneous—evidence for improving sleep quality. Nevertheless, caution is warranted when generalizing these findings to the broader population, as the included studies differed considerably in terms of age groups, clinical settings, and study methodologies. [18]

4.7. Nasal Strips

Nasal dilator strips are used to mechanically widen the nasal passages in order to increase airflow and thereby facilitate breathing during sleep. However, whether this mechanism translates into clinically meaningful improvements in sleep quality or sleep apnea is a separate question. [19]

A study involving 54 pregnant women with sleep-disordered breathing found no significant differences between the use and non-use of nasal dilator strips in terms of the respiratory event index or other objective respiratory measures. Subjective sleep outcomes also did not differ between the two conditions. [19]

These findings do not support the conclusion that nasal strips produce clinically meaningful improvements in sleep apnea or sleep quality, at least within the population studied. However, the generalizability of these results is limited because the study was restricted to pregnant women with elevated body mass index and a history of habitual snoring. [19]

4.8. Smart Beds

Smart beds are among the technologies designed to optimize sleep by monitoring various parameters during the night, including body movement, heart rate, respiratory rate, and sleep timing. However, controlled clinical evidence demonstrating that the information provided by these devices directly improves sleep health remains limited. [20,21]

A large-scale observational study analyzing data from more than 330,000 smart bed users reported that individuals with regular sleep schedules experienced more restorative sleep and had lower average heart and respiratory rates than those with irregular sleep patterns. However, because the study was observational, it cannot be concluded that the smart bed itself caused these outcomes; rather, it demonstrates an association between regular sleep patterns and better sleep and cardiorespiratory measures. [20]

Another review of sleep monitoring technologies likewise concluded that these technologies have considerable potential for sleep assessment but continue to face important limitations regarding measurement accuracy, data interpretation, and performance across different individuals. [21]

Therefore, although smart beds and similar sleep technologies are promising tools for the future of sleep optimization, they should not currently be classified as interventions that have been proven to improve sleep. Existing evidence primarily supports their use for monitoring sleep behaviors and evaluating specific sleep characteristics rather than directly enhancing sleep quality. [20,21]

4.9. Overall Level of Evidence for Sleepmaxxing Interventions

When the current body of evidence is considered collectively, it is evident that the scientific support for interventions used within the sleepmaxxing framework varies considerably. Some interventions, such as melatonin and white noise, have demonstrated beneficial effects in specific populations or for particular sleep outcomes; however, these findings cannot be generalized to the broader population in all circumstances. [14,18]

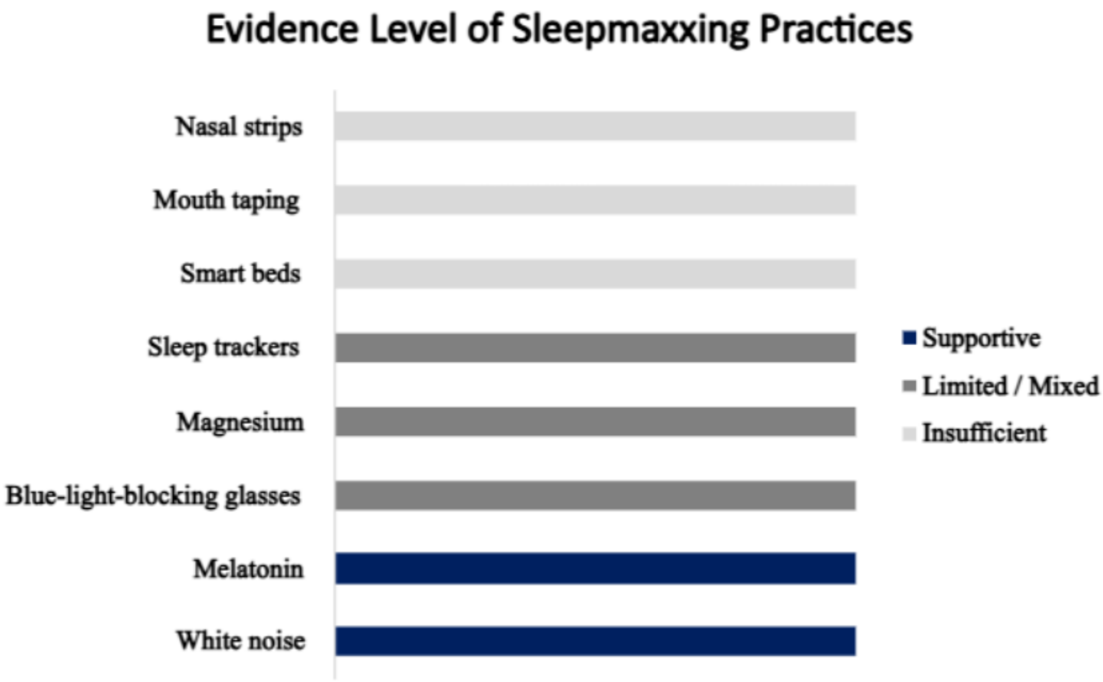
Although blue light-blocking glasses and magnesium supplementation are supported by biologically plausible mechanisms and some positive research findings, the results of randomized controlled trials remain inconsistent. In particular, the most recent meta-analysis of randomized controlled trials found no significant effects of blue light-blocking glasses on objective actigraphy-based sleep outcomes, while for magnesium, a clear discrepancy exists between observational studies and randomized controlled trials. [8,12,13]

For interventions such as mouth taping and nasal strips, positive findings have been reported in selected patient populations; however, there is currently insufficient evidence to conclude that these methods improve sleep quality in the general population. The discrepancy between social media claims and the available scientific evidence is particularly notable in the case of mouth taping. [10,11,19]

Consumer sleep trackers and smart beds, on the other hand, should be regarded as technologies designed to measure and monitor sleep behavior rather than as direct therapeutic interventions. The measurement accuracy of these devices varies substantially across products and does not consistently correspond with reference methods such as polysomnography. Therefore, a sleep score generated by a consumer device should not be considered a definitive indicator of an individual's actual sleep quality. [16,17,20,21]

Overall, the scientific evidence supporting sleepmaxxing interventions is heterogeneous; some strategies are supported by evidence, whereas others are based on limited or insufficient data. Consequently, presenting all of these methods together as a single "sleep optimization" package on social media is scientifically misleading. [8–21]

4.10. Summary of the Evidence



Graph 3. Classification of Sleepmaxxing Practices According to Their Level of Scientific Evidence.

The distribution of the evidence levels for the practices evaluated within the scope of Sleepmaxxing is summarized in Graph 3. These findings suggest that, rather than representing a scientifically validated standalone “sleep optimization protocol,” Sleepmaxxing is a heterogeneous collection of sleep-related behaviors, dietary supplements, and consumer technologies supported by varying levels of scientific evidence. [8–21]

5. Current Debates and Knowledge Gaps

With the increasing visibility of sleep optimization practices through social media, Sleepmaxxing has evolved beyond an individual sleep hygiene approach to become part of the broader digital health culture. In particular, meta-analytic evidence demonstrating associations between social media use and depression, anxiety, and sleep problems among young people highlights the importance of considering the digital environment in which sleep optimization content is both created and consumed. Nevertheless, although current research has established a relationship between social media use and sleep, studies directly investigating the Sleepmaxxing phenomenon itself—how such content shapes individuals' sleep behaviors and why it has gained widespread popularity—remain very limited. [22]

5.1. Social Media, Screen Use, and Sleep Optimization

While social media facilitates the dissemination of information and recommendations related to sleep health, it may also contribute to the development of unrealistic expectations regarding sleep. Social media use—particularly problematic social media use—has been positively associated with depression, anxiety, and sleep disturbances, although substantial heterogeneity has been observed across studies. These findings suggest that, rather than viewing social media use as inherently harmful, its effects should be considered in the context of usage patterns, content type, and individual characteristics. [22]

Studies using objective measures of pre-sleep screen use further indicate that this relationship is more complex than commonly assumed. Among adolescents, total screen time during the two hours before bedtime was not significantly associated with most sleep outcomes. In contrast, screen use while in bed, interactive screen activities, gaming, and multitasking were associated with shorter sleep duration or longer sleep onset latency. These findings suggest that, instead of recommending the complete avoidance of all screen use before bedtime, greater emphasis should be placed on the timing, content, and level of interactivity of screen-based activities. [23]

5.2. The Paradox of Sleep Optimization: Orthosomnia and Sleep Anxiety

One of the fundamental concerns associated with the Sleepmaxxing approach is that the pursuit of better sleep may gradually evolve into excessive monitoring and control of sleep itself. This phenomenon has been increasingly discussed under the concept of orthosomnia, particularly with the widespread use of consumer sleep-tracking devices. In a study involving 523 participants from the general population, the prevalence of orthosomnia ranged from

3.0% to 14.0%, depending on the diagnostic criteria applied, and 35.8% of participants reported regularly using sleep-tracking devices. Individuals with orthosomnia exhibited greater sleep-related anxiety, obsessive thoughts about sleep quality, and more severe insomnia symptoms, suggesting that continuous monitoring of sleep data does not necessarily result in better sleep. [24] The paradoxical optimization cycle that may emerge through this process is conceptually illustrated in Figure 3.

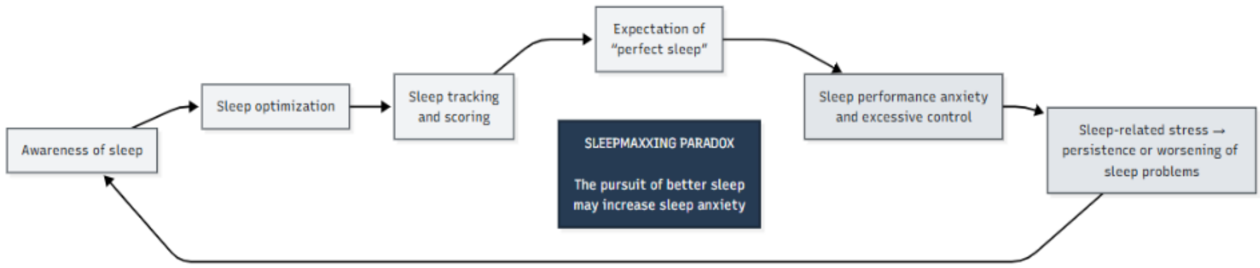


Figure 3. The Paradoxical Cycle of Sleep Optimization in the Sleepmaxxing Approach.

This highlights one of the fundamental paradoxes of Sleepmaxxing: behaviors intended to improve sleep health may, if they foster excessive control over sleep and unrealistic performance expectations, ultimately increase sleep-related anxiety. The relationship between sleep and anxiety is bidirectional, with insufficient or disturbed sleep contributing to increased anxiety, while anxiety itself can perpetuate sleep disturbances. Consequently, sleep optimization should not be viewed solely as a matter of "more data, more monitoring, and more intervention." Instead, consideration should also be given to an individual's need for control over sleep and the associated psychological burden. [24,27]

5.3. The Quality of Sleep Information and the Problem of Misinformation

The role of social media in the spread of Sleepmaxxing extends beyond the sharing of behaviors; the way sleep-related information is produced and validated also represents a significant concern. The dissemination of inaccurate or misleading information about sleep through social media, recommendations from non-experts, and commercial interests may lead individuals to misunderstand evidence-based sleep practices. More importantly, sleep deprivation has been suggested to impair cognitive processes such as memory, emotional regulation, and decision-making, potentially making individuals more susceptible to misinformation. As a result, a self-reinforcing cycle may emerge in which misinformation contributes to poor sleep, while poor sleep further increases vulnerability to misinformation. [26]

In this context, sleep-tracking devices and artificial intelligence–based applications present a dual potential. On the one hand, they may improve sleep literacy by providing individuals with information about their sleep. On the other hand, they may also perpetuate misinformation through inaccurate algorithms, commercially driven recommendations, or advice that has not been sufficiently validated by scientific evidence. Therefore, future research should focus not only on the question of "Which Sleepmaxxing practices are effective?" but also on where these practices are learned, the extent to which they are supported by scientific evidence, and how they influence individuals' sleep-related behaviors. [26]

5.4. Current Knowledge Gaps

The current literature indicates that many of the mechanisms associated with Sleepmaxxing have not yet been directly investigated. Although growing evidence exists regarding the relationships among social media use, screen exposure, sleep tracking, sleep-related anxiety, and mental health, studies that directly define and measure the Sleepmaxxing phenomenon remain limited. Furthermore, because many existing studies employ cross-sectional designs, it is not yet possible to determine whether social media use or sleep optimization behaviors are causes, consequences, or components of a bidirectional relationship with sleep and mental health problems. [22,24,27]

Similarly, stronger and longer-term research is needed to evaluate the effectiveness of sleep-related interventions. For example, a small randomized pilot trial involving 40 participants investigating PS128 reported improvements in certain depressive symptoms, fatigue, and sleep parameters; however, these findings require confirmation in larger studies with longer follow-up periods. [25] Therefore, future Sleepmaxxing research should move beyond simply increasing sleep duration or improving sleep scores and instead examine the psychological consequences of sleep optimization, the risk of orthosomnia, social media–driven misinformation, and the long-term behavioral effects of these practices. [22,24,25,26]

Overall, the current debate surrounding Sleepmaxxing is shifting from the question of "What should people do to sleep better?" toward "At what point does the pursuit of better sleep become excessive optimization?" Existing evidence suggests that social media, screen use, sleep tracking, anxiety, and misinformation may each represent different components of this process. However, the limited number of studies directly investigating Sleepmaxxing indicates that the long-term consequences of this emerging digital sleep culture remain largely uncertain. [22,23,24,26,27]

6. Clinical and Public Health Implications

From a clinical perspective, the key consideration in evaluating sleep optimization trends such as Sleepmaxxing is to assess sleep quality, regularity, daytime functioning, and existing sleep complaints comprehensively, rather than focusing solely on sleep duration or a single sleep score. It has been suggested that conventional measures used in clinical sleep assessment may, in some cases, fail to adequately explain the underlying causes of a patient's complaints, and that evaluating sleep quality using a single metric is inherently limited. Therefore, isolated indicators such as the "sleep score" commonly used in Sleepmaxxing should not be considered substitutes for comprehensive clinical assessments. [28]

With regard to sleep hygiene, it is important not only to provide information but also to understand how individuals perceive this information and incorporate it into their daily lives. A study conducted among perimenopausal women demonstrated that knowledge of sleep hygiene and sleep disorders was associated with attitudes and practices, and that better knowledge and more positive attitudes contributed to the adoption of healthier sleep behaviors. These findings suggest that, within the context of Sleepmaxxing, teaching evidence-based and practical sleep behaviors may be more beneficial than simply presenting individuals with numerous sleep optimization recommendations. [29]

Digital sleep technologies and wearable devices offer significant opportunities for the long-term and objective monitoring of sleep health. A large-scale study demonstrated that long-term sleep data collected from commercial wearable devices can provide valuable information on sleep duration, sleep regularity, and sleep stages. In particular, sleep irregularity was found to be associated with several chronic conditions, including obesity, hypertension, hyperlipidemia, major depressive disorder, and generalized anxiety disorder. Therefore, when used appropriately, these technologies may serve as valuable tools for both individual awareness and public health research. [30]

Nevertheless, the interpretation of sleep technology data is crucial in clinical practice. Sleep tracking can provide useful insights into an individual's sleep health; however, translating a score generated by a consumer device alone into a definitive clinical conclusion such as "I slept well" or "I slept poorly" is not appropriate. Particularly within the context of Sleepmaxxing, this approach may encourage continuous monitoring of sleep data and promote the perception of sleep as a performance metric. Therefore, in the clinical use of digital sleep technologies, generating data is only one aspect; equally important is the accurate interpretation of these data and, when necessary, their integration with professional clinical evaluation. [28,30]

From a public health perspective, integrating sleep education more extensively into the training of healthcare professionals is also essential. In a qualitative study involving social work students, participants expressed a desire for more education on the importance of sleep health, sleep deprivation and the signs of sleep disorders, lifestyle and environmental factors affecting sleep, and behaviors that promote healthy sleep. These findings suggest that sleep health should not be regarded solely as the responsibility of sleep specialists, but rather as a fundamental area of knowledge that should be shared across a wide range of healthcare professionals. [31]

The importance of sleep health from a public health perspective is also evident among populations exposed to demanding working conditions, such as healthcare professionals. The association between poor sleep quality and overweight or obesity among healthcare workers demonstrates that sleep health is not merely an individual lifestyle choice but a broader public health issue closely linked to both physical and mental well-being. Therefore, health communication regarding Sleepmaxxing should aim not to encourage greater sleep tracking or more intensive "optimization" behaviors, but rather to promote evidence-based, sustainable sleep practices that take an individual's overall health into account. [32]

In conclusion, from both clinical and public health perspectives, Sleepmaxxing should be viewed less as a direct therapeutic approach and more as a reflection of the growing awareness of sleep health in the digital age. When evaluating this trend, healthcare professionals should consider the sleep technologies used by individuals, their knowledge and attitudes regarding sleep, their actual sleep complaints, and their daytime functioning as a whole, while ensuring that digital sleep data do not replace comprehensive clinical assessment. [28–32]

7. Future Research Directions

A major research gap in the scientific evaluation of Sleepmaxxing behaviors is the lack of standardized and objective methods for measuring sleep optimization practices. The current literature emphasizes the need for objective measurement tools to assess sleep timing, sleep regularity, and environmental factors, and suggests that the development of accessible and reliable methods for evaluating sleep and circadian physiology could substantially advance this field of research. Therefore, future studies should prioritize the development of valid and reliable instruments capable of directly assessing Sleepmaxxing behaviors and their effects on sleep health. [33]

Wearable technologies provide an important opportunity to investigate sleep behaviors over extended periods under real-world conditions. However, while current devices generally perform well in estimating sleep duration and timing, their accuracy in identifying specific sleep stages remains more limited. Consequently, future research should compare wearable devices with gold-standard methods such as polysomnography in larger populations and over longer follow-up periods. In addition, further studies are needed to evaluate the measurement accuracy of different devices and to investigate how individual characteristics influence device performance. [34]

Future research on Sleepmaxxing should evaluate not only the short-term outcomes of sleep optimization interventions but also their long-term effects. Large randomized controlled trials of insomnia treatments have demonstrated that although some interventions are effective in the short term, long-term evidence regarding their efficacy and safety remains limited for many approaches. Therefore, future longitudinal and randomized controlled studies should investigate the effects of Sleepmaxxing behaviors on sleep quality, mental health, sleep-related anxiety, and overall health. Such studies should also aim to determine which sleep optimization behaviors provide genuine benefits and at what point these practices transition into excessive control or unnecessary intervention. [35]

8. Conclusion

Sleepmaxxing can be regarded as an emerging approach that has gained prominence in recent years through social media and digital health culture, aiming to optimize sleep duration, quality, regularity, and environmental conditions using a combination of behavioral, technological, and biological strategies. However, the current scientific literature indicates that Sleepmaxxing is neither a single, comprehensive intervention nor a scientifically validated standardized "sleep optimization protocol." Rather, it is a heterogeneous concept that encompasses numerous practices supported by varying levels of scientific evidence. Therefore, the scientific evaluation of Sleepmaxxing should focus not on its popularity on social media but on the evidence supporting each individual practice.

Current evidence suggests that several core behaviors promoted within Sleepmaxxing are consistent with established principles of healthy sleep. Maintaining adequate and regular sleep duration, preserving a consistent sleep–wake schedule, engaging in appropriate physical activity, and following healthy dietary habits all play important roles in promoting sleep health. Nevertheless, these findings do not imply that "more sleep" or "more optimization" invariably leads to better outcomes. In particular, when sleep problems reach a clinical level, it appears more appropriate to assess the underlying cause and, when indicated, implement evidence-based interventions such as cognitive behavioral therapy for insomnia (CBT-I), rather than simply attempting to increase sleep duration or improve sleep scores generated by consumer devices.

The evidence regarding the products and technologies commonly associated with Sleepmaxxing presents a more complex picture. Although melatonin and white noise may produce beneficial effects under specific conditions and for certain sleep outcomes, these findings cannot be directly generalized to the general population. Similarly, despite biologically plausible mechanisms and some positive findings, current randomized controlled trials do not provide consistent or conclusive evidence supporting the effectiveness of blue-light-blocking glasses or magnesium supplementation. For highly popular social media practices such as mouth taping and nasal strips, limited positive findings have been reported in selected populations; however, there is insufficient evidence to conclude that these interventions improve sleep quality in the general population. Likewise, although consumer sleep trackers and smart beds offer valuable opportunities for monitoring sleep behaviors, it remains essential to distinguish between the ability to measure sleep and the ability to genuinely improve it.

At this point, one of the most important paradoxes of Sleepmaxxing becomes apparent: optimization behaviors initiated with the goal of improving sleep may ultimately increase sleep-related anxiety rather than promote sleep health when they evolve into excessive monitoring, control, and performance expectations. In particular, the continuous tracking of sleep scores generated by wearable sleep devices and the pursuit of "perfect sleep" may contribute to behaviors resembling orthosomnia. Therefore, healthy sleep should not be regarded solely as a measurable indicator of biological performance but should instead be evaluated in conjunction with an individual's daytime functioning, subjective well-being, sleep-related anxiety, and overall life circumstances.

The dissemination of Sleepmaxxing through social media also represents an important public health consideration. While social media facilitates the widespread dissemination of evidence-based information about sleep, it also enables practices supported by limited scientific evidence to be presented as definitive and universally effective solutions. Consequently, future health communication should focus not on encouraging individuals to adopt more products, technologies, or "sleep hacks," but rather on helping them access accurate information, critically evaluate the available evidence, and seek professional medical advice when appropriate.

In conclusion, Sleepmaxxing should not be viewed as a trend that ought to be entirely rejected on scientific grounds but rather as a reflection of the growing public interest in sleep health within the digital era. However, distinguishing between its genuinely beneficial components and those supported by limited evidence or associated with potential risks is of critical importance. Current evidence indicates that sleep health is not synonymous with "more monitoring, more intervention, or higher sleep scores." Instead, it depends on maintaining sufficient and regular sleep, adopting sustainable lifestyle habits, optimizing environmental conditions, and implementing evidence-based clinical interventions when necessary. Future research priorities should include the development of valid and reliable instruments for directly measuring Sleepmaxxing behaviors, the investigation of their long-term clinical and psychological outcomes, improvements in the accuracy of wearable sleep technologies, and the examination of the impact of social media–driven misinformation. Such efforts may allow Sleepmaxxing to evolve beyond a popular digital trend into a meaningful area of scientific inquiry that enhances our understanding of the relationship between sleep health and individual optimization.

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Conflict of Interest Statement

No conflict of interest has been declared.